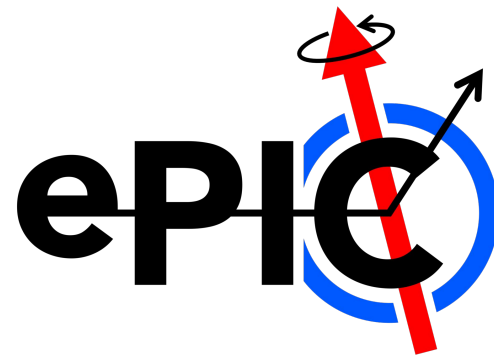


Optimization of machine learning methodology for D0 analysis at the EIC

Connie Yang, Deepa Thomas
April 28, 2026



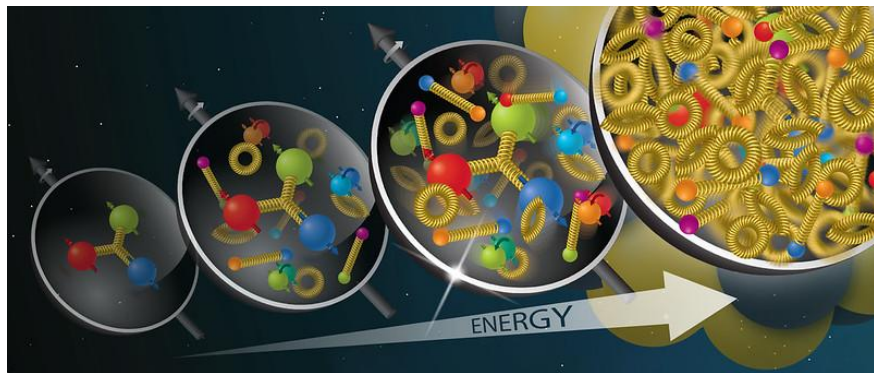
Presentation outline

- Background and motivation
- Preliminary methods and results from Fall 2025
- Investigations to improve methodology
- Comparison of methods to non-ML based analysis

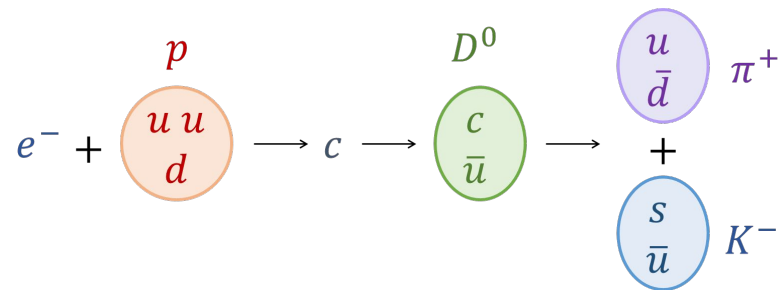
Background and motivation

Heavy flavor physics at the EIC

- Electron-Ion Collider (EIC) will image the proton and reveal gluon dynamics at high energy
- D0 mesons serve as heavy flavor probes to access the gluons

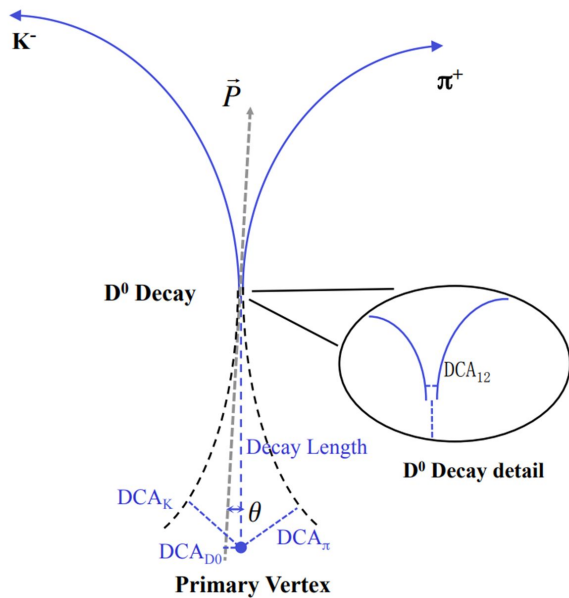


<https://www.bnl.gov/eic/goals.php>

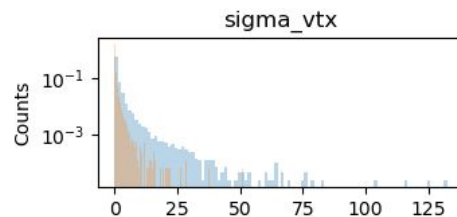
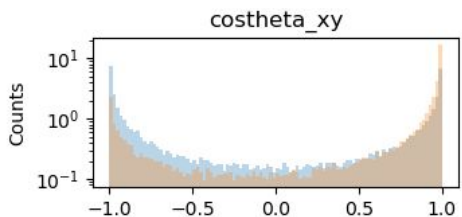
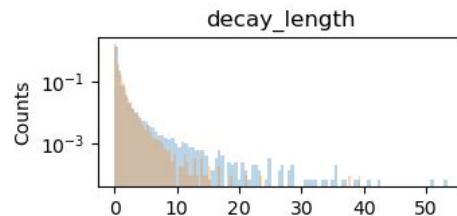
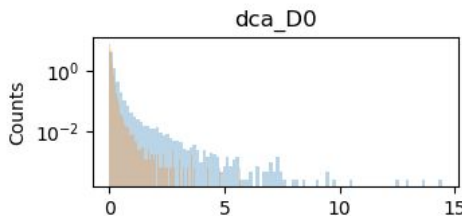
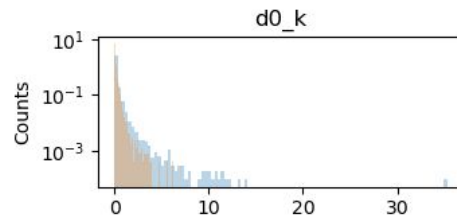
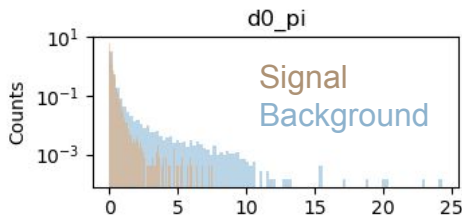


Project goal

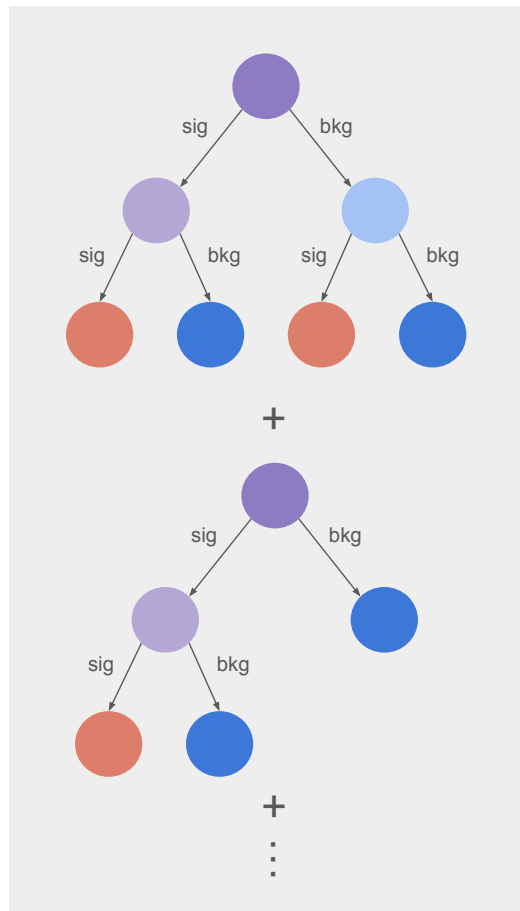
Use boosted decision trees (BDTs) to separate D0 signals from background based on decay topology.



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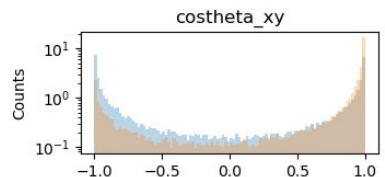
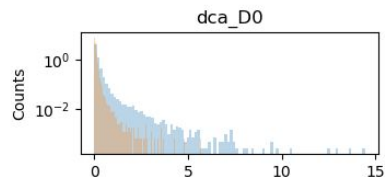
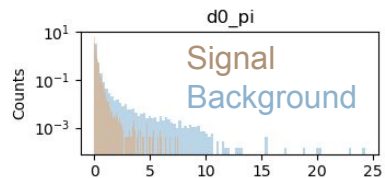
Boosted decision tree (BDT) based analysis



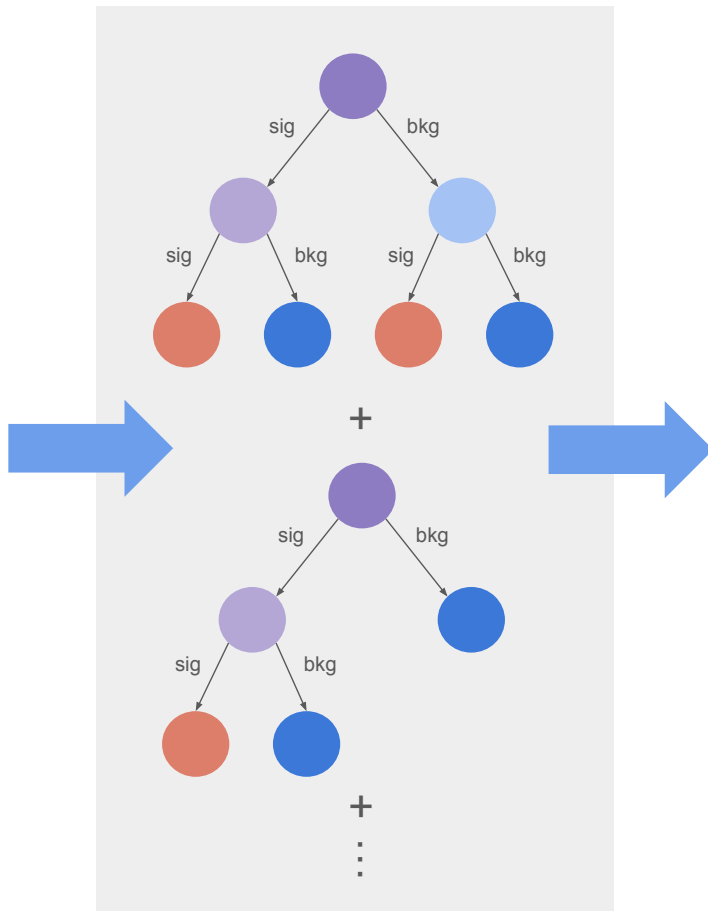
- Decision trees classify candidates as sig or bkg by applying cuts on topological features.
- Boosting involves training subsequent trees to correct the errors of previous trees.
- Overall BDT can achieve better performance and stability.
- Model structure is controlled by hyperparameters (number of trees, depth of each tree, etc.).

Boosted decision tree (BDT) based analysis

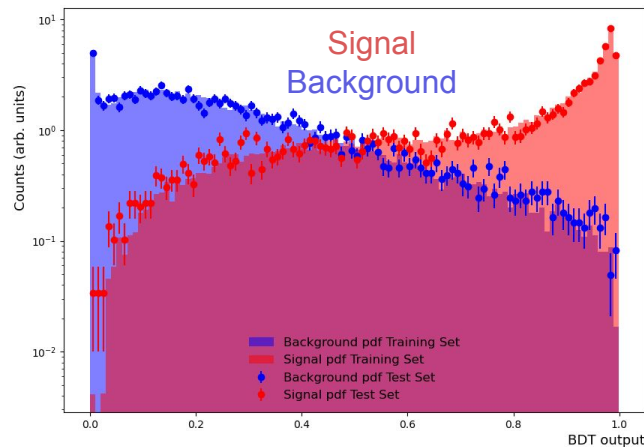
BDT is trained on the topological features of sig+bkg candidates.



⋮



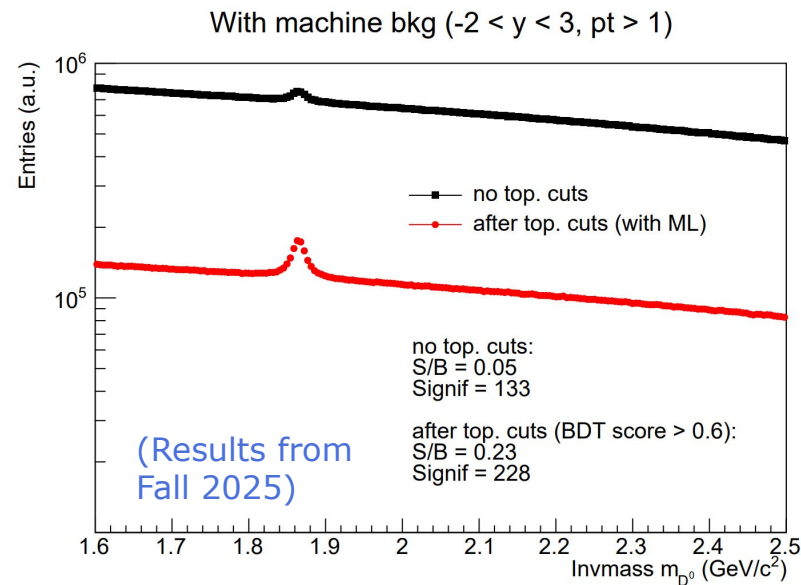
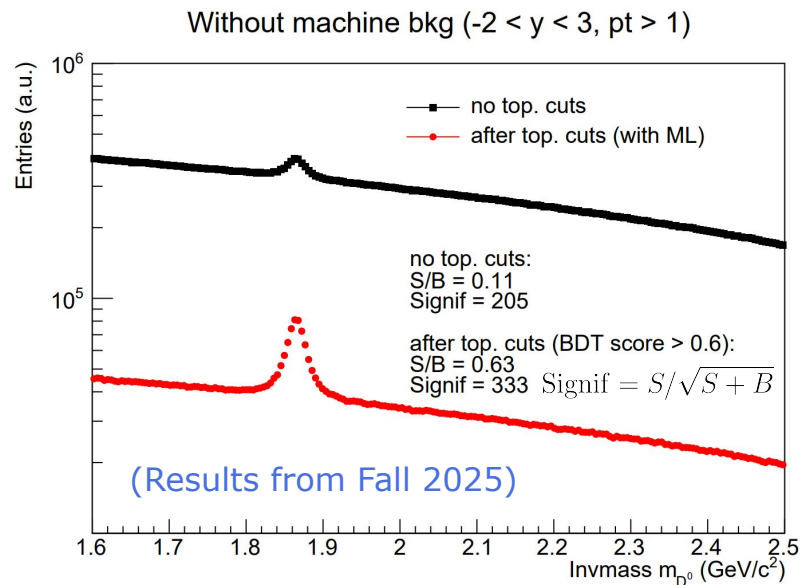
BDT outputs the likelihood that a candidate is signal or background.



→ Multidimensional problem reduces to a single variable.

Preliminary analyses

Study of machine-background effects



Fall 20225: Applied existing BDT framework to analyze the impacts of machine-background effects (synchrotron radiation and beam particle interactions)

Spring 2026: Investigate ways to improve BDT framework (for samples without machine-background effects)

Improvements to methodology

Task I: Train BDT separately for different y and p_T slices

Task II: Examine the impacts of various feature selections

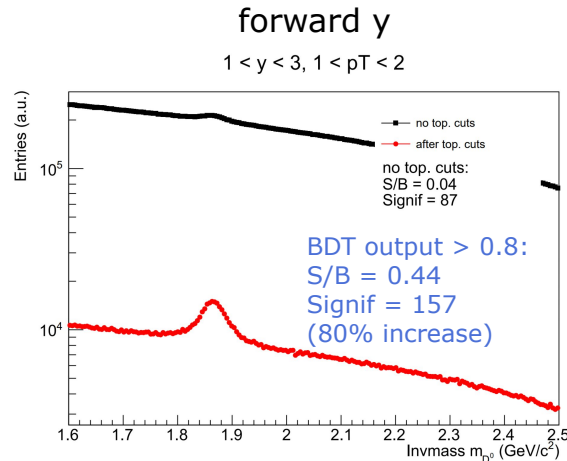
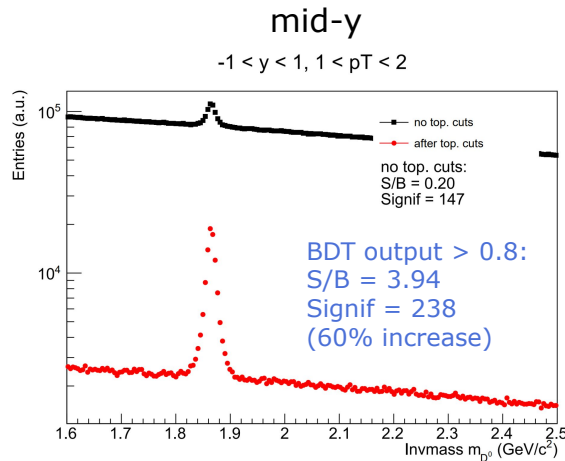
Task III: Tune BDT hyperparameters

Task I:

Analysis in various y and pT

D0 invariant mass in various y and p_T slices

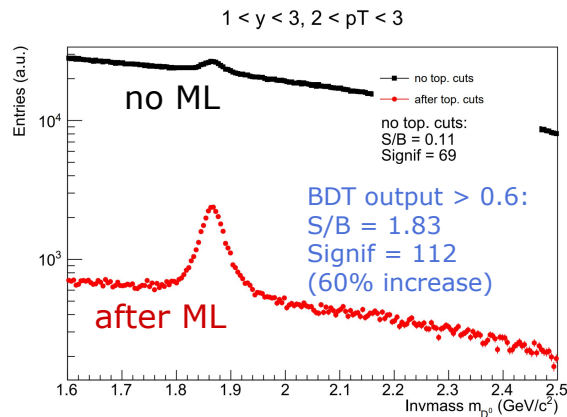
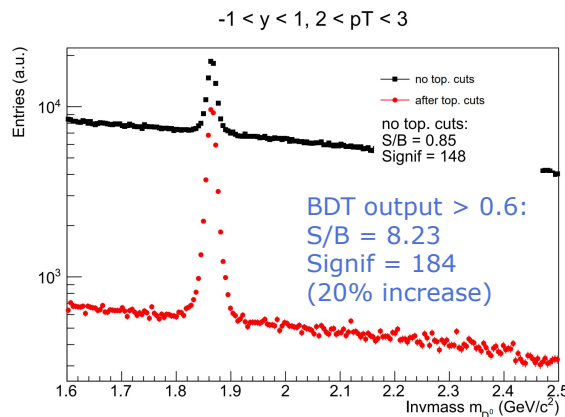
lower p_T



Same features and hyperparameters as preliminary analyses.

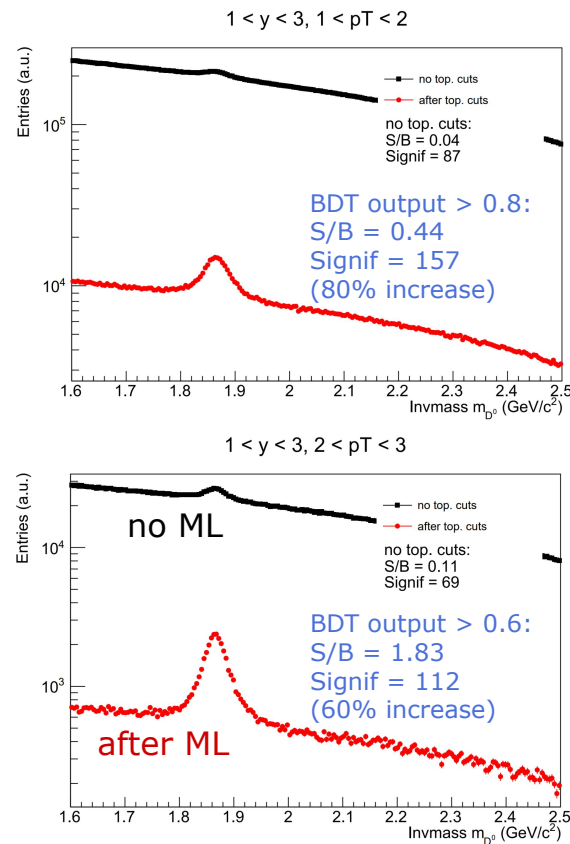
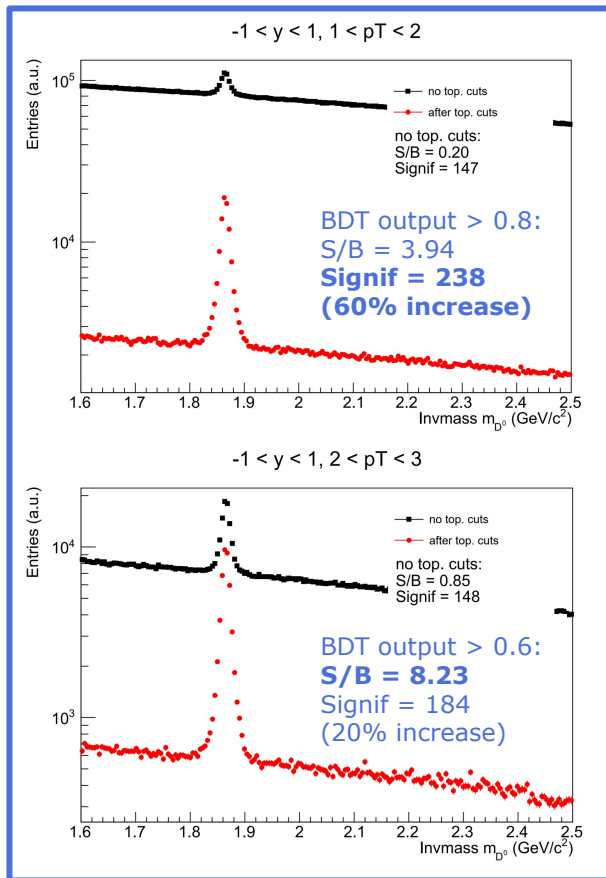
Trained separate BDT for each slice.

higher p_T



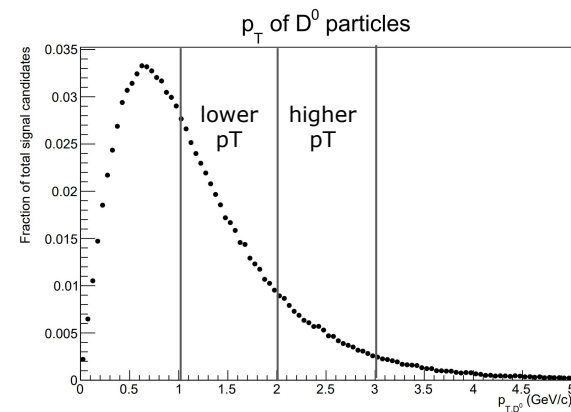
Optimized each BDT threshold by maximizing Signif.

D0 invariant mass: lower pT vs. higher pT

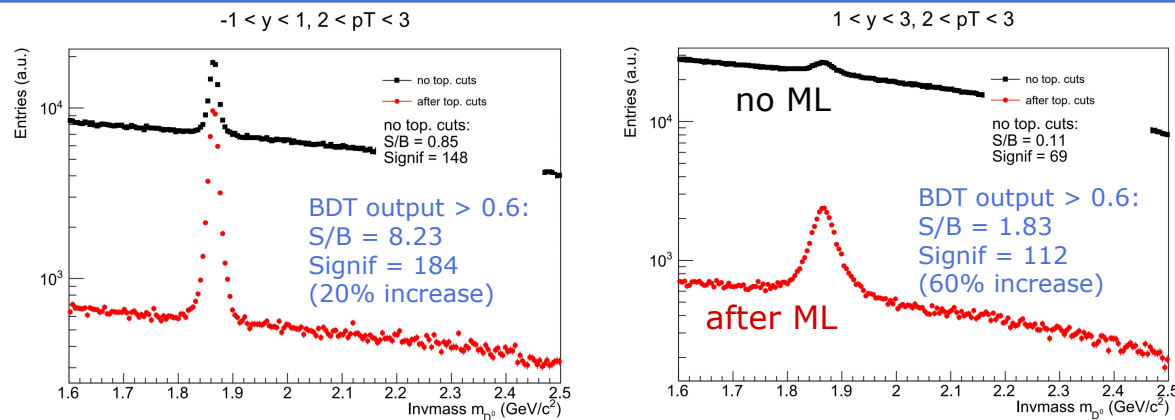
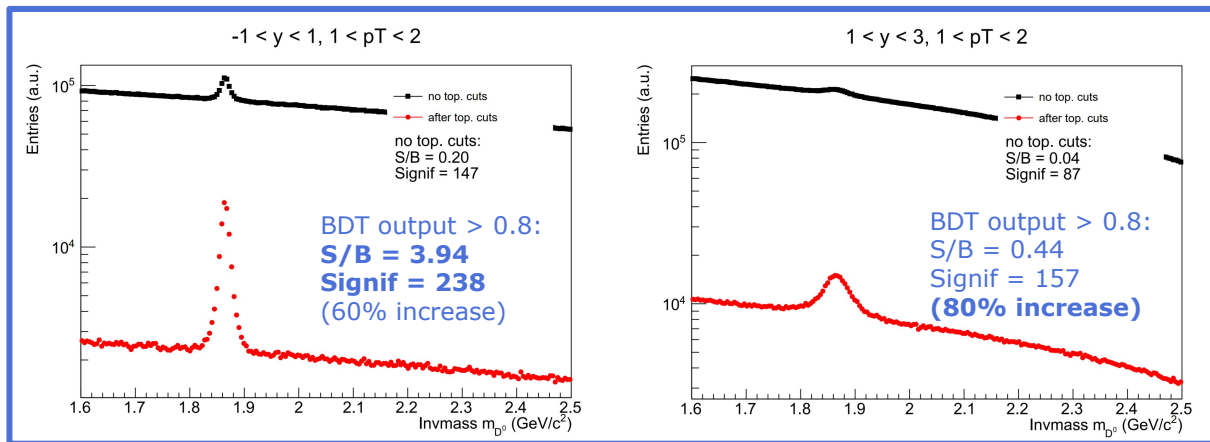


At lower pT, S/B is lower but Signif is higher.

- Worse reconstruction
- Better statistics
- Greater improvement in Signif



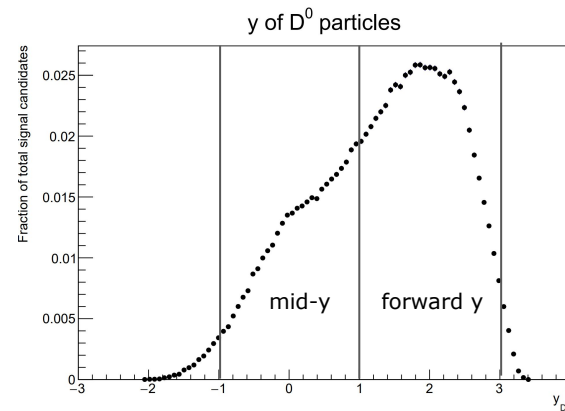
D0 invariant mass: mid-y vs. forward y

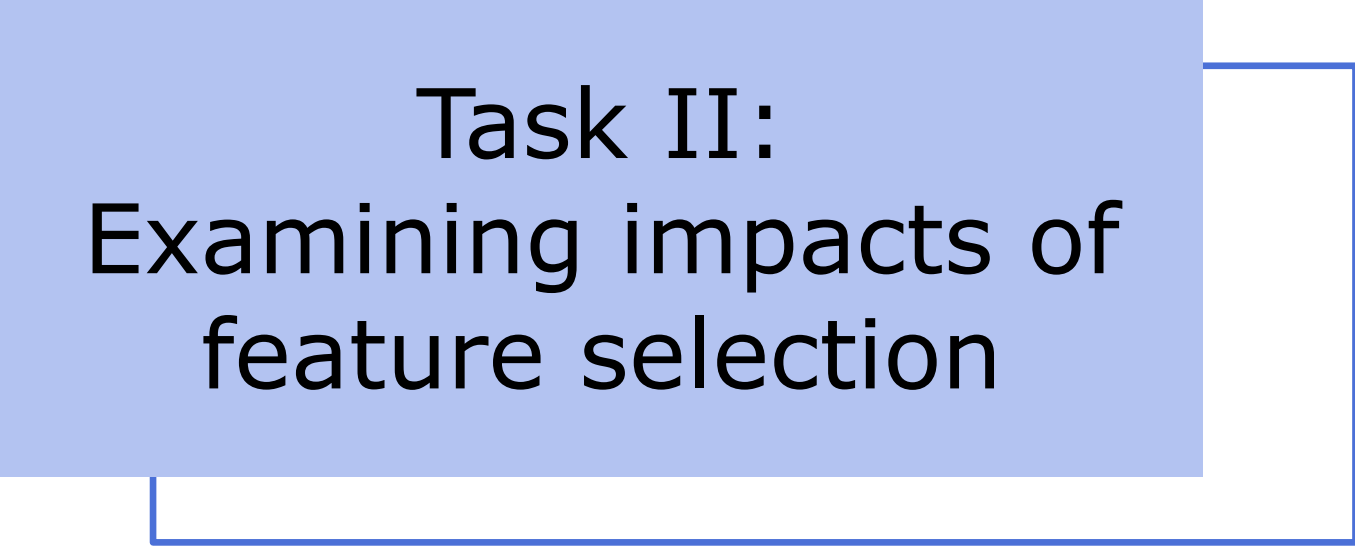


At forward y , results are worse but improvement in Signif is greater.

- Large background
- Worse reconstruction

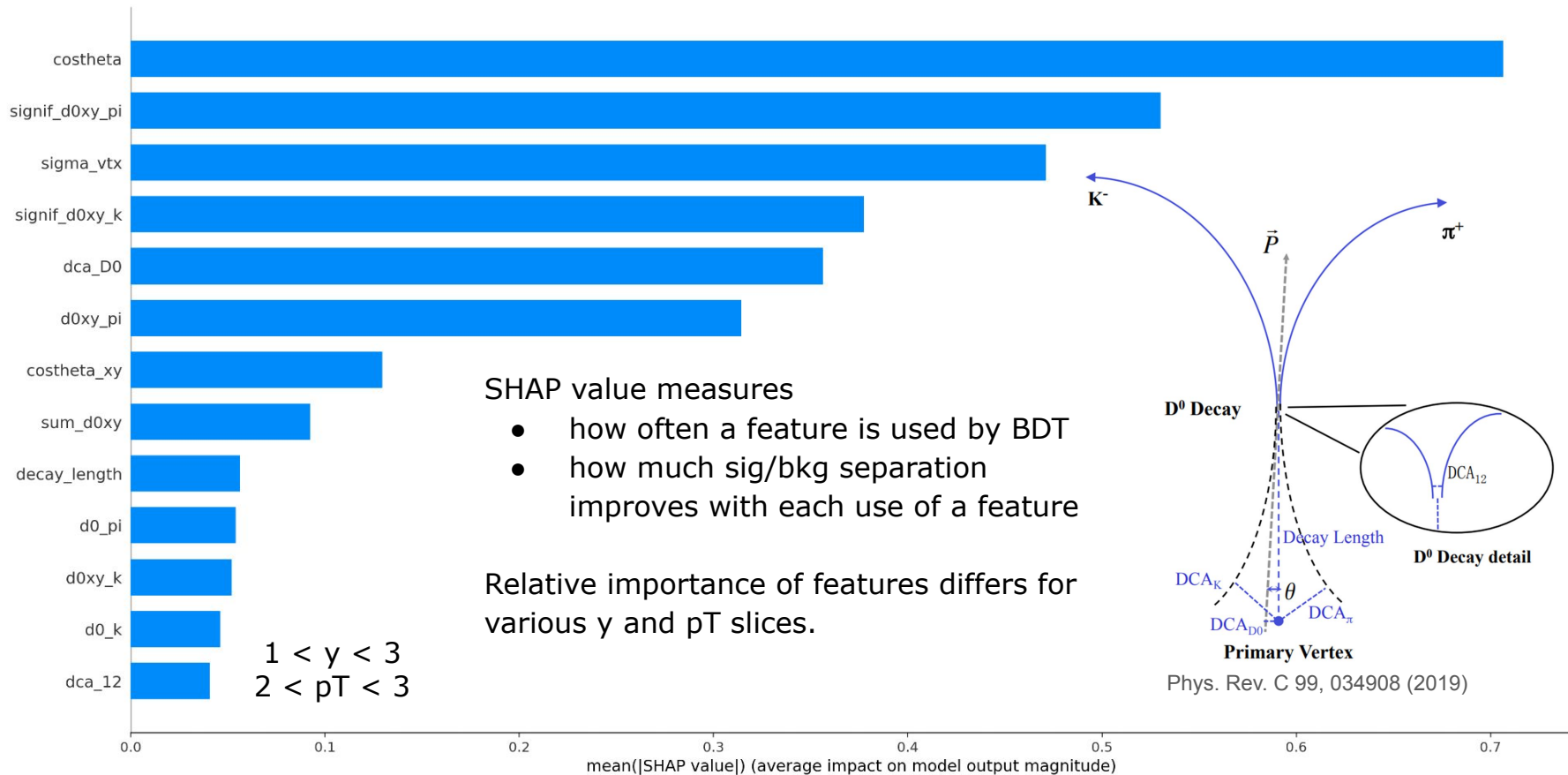
→ Greatest impact in forward y and lower p_T .





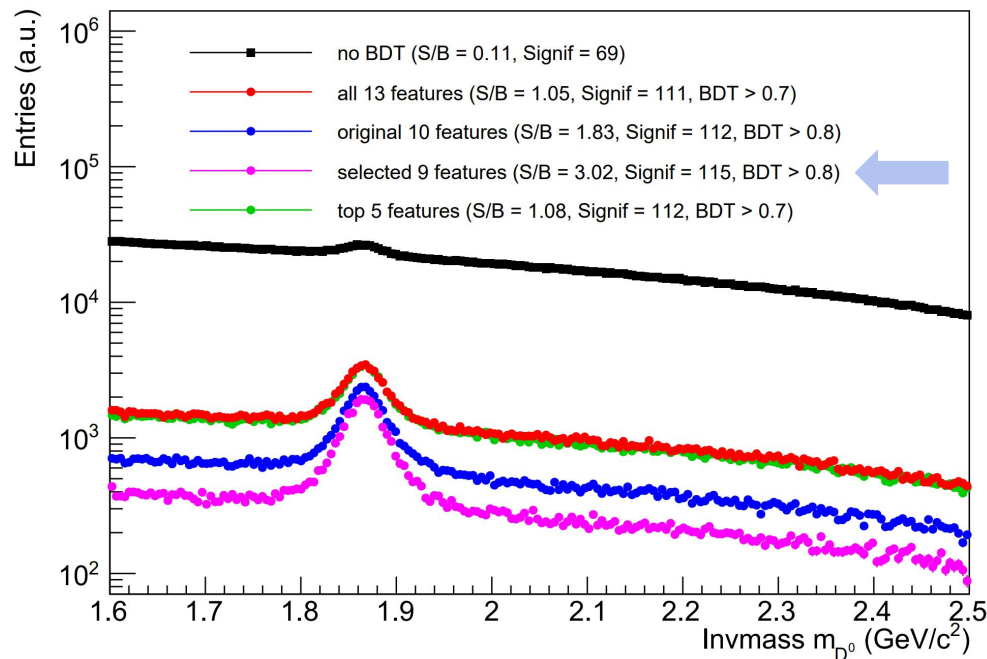
Task II: Examining impacts of feature selection

Relative importance of topological features



Results for various feature combinations

$1 < y < 3, 2 < p_T < 3$



All 13 features:

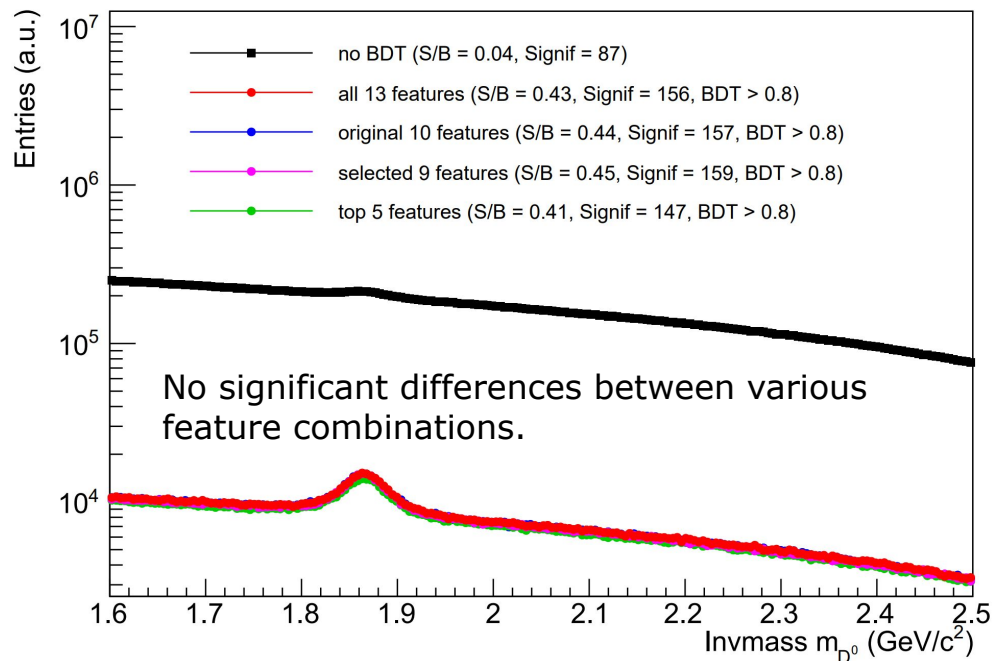
- top 5
- costheta
 - sigma_vtx
 - signif_d0xy_pi
 - signif_d0xy_k
 - dca_D0
 - costheta_xy
 - decay_length
 - d0_pi
 - d0_k
- selected 9 (best combination)
- d0xy_pi
 - d0xy_k
 - sum_d0xy
 - dca_12

Adding more features does not necessarily improve BDT.

- Additional features may have low importance.
- Too many features can lead to overfitting.

Results for various feature combinations

$1 < y < 3, 1 < p_T < 2$



Impact of feature selection varies in different y and p_T slices.

- BDT is less sensitive to feature selection at low p_T due to larger statistics.
- Feature selection is more important for physical and practical considerations.

Task III: Tuning BDT hyperparameters

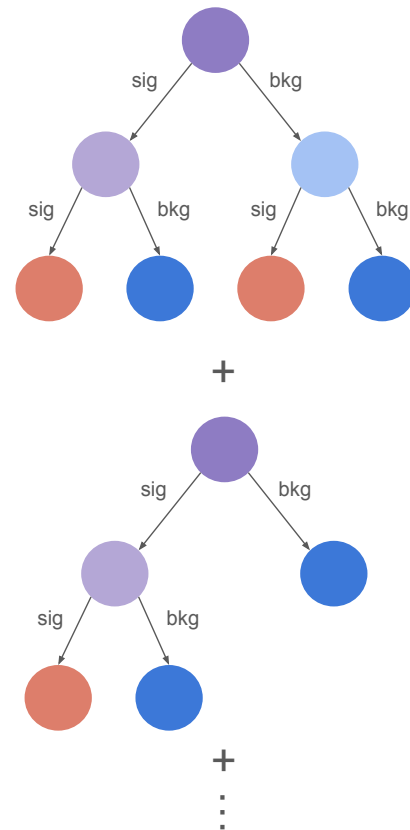
BDT hyperparameters examined

- `n_estimators`: number of trees
- `max_depth`: max number of iterations per tree
- `learning_rate`: size of each boosting step

Basic hyperparameters

- `min_child_weight`: min number of candidates required in each node
 - Default: 1
- `min_split_loss`: min improvement required to split a node
 - Default: 0

Regulators to help prevent overfitting



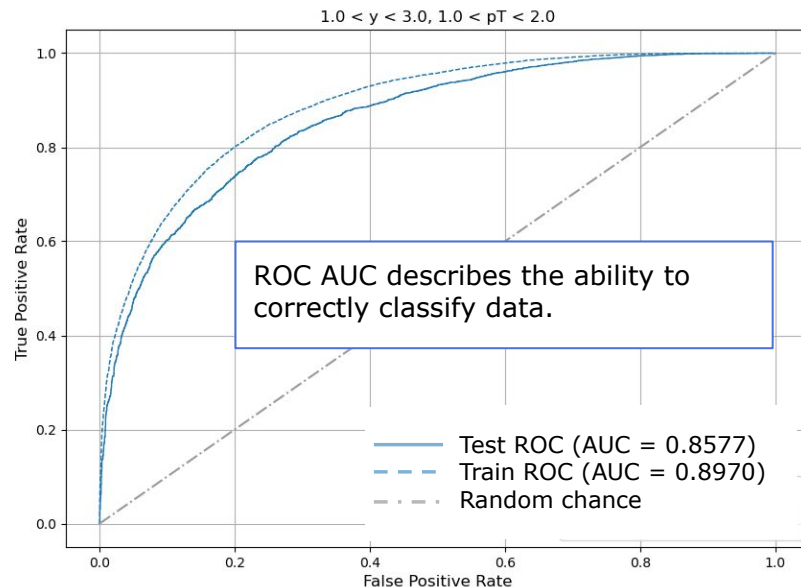
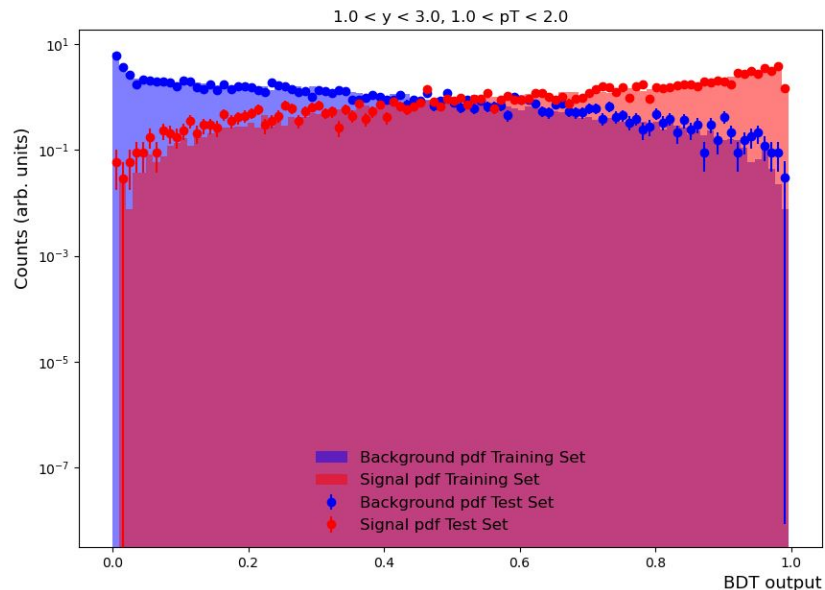
Optuna tests combinations of hyperparameters to find the best values.

BDT results: basic hyperparameter optimization

n_estimators: (100, 500) → 480
max_depth: (1, 3) → 3
learning_rate: (0.01, 0.1) → 0.067
min_child_weight: default → 1
min_split_loss: default → 0

BDT is well-trained and stable.

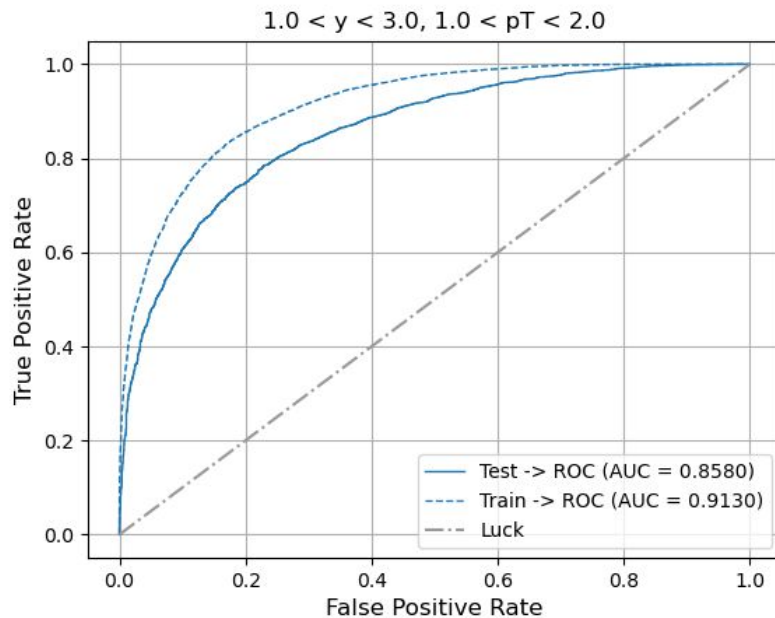
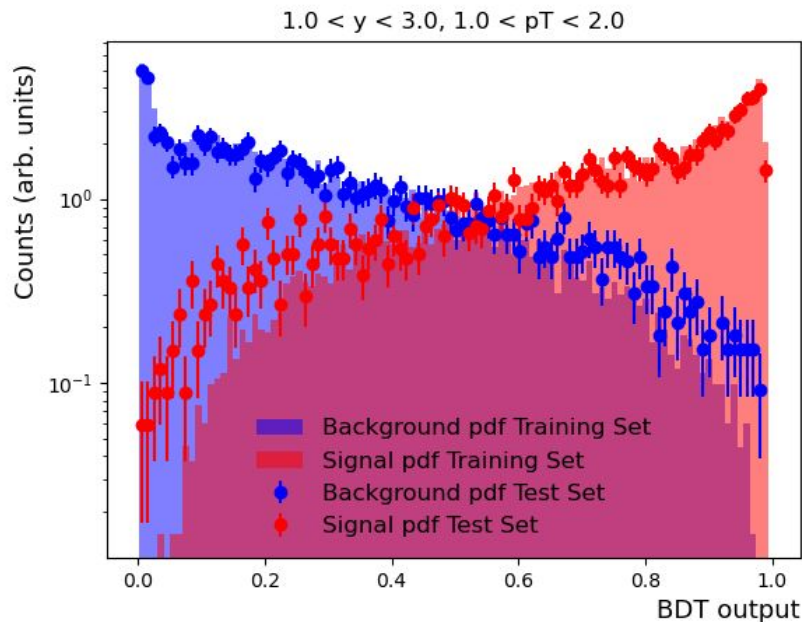
- Good ROC AUC
- Similar results for training and test data



BDT results: increased hyperparameter flexibility

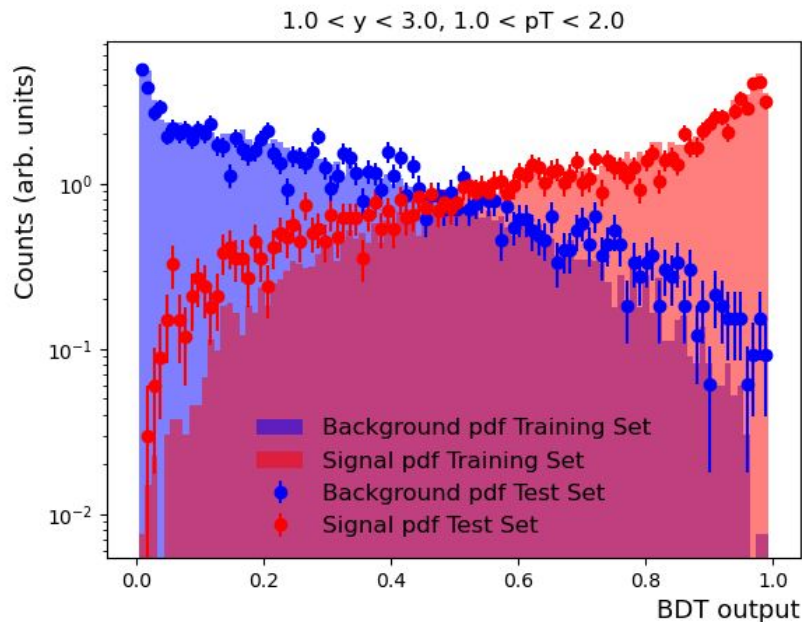
n_estimators: (100, 800) $\rightarrow$ 347
max_depth: **(1, 8)** $\rightarrow$ **6**
learning_rate: (0.01, 0.3) $\rightarrow$ 0.034
min_child_weight: default $\rightarrow$ 1
min_split_loss: default $\rightarrow$ 0

- No significant improvement in test ROC AUC
- Overfitting due to large tree depth



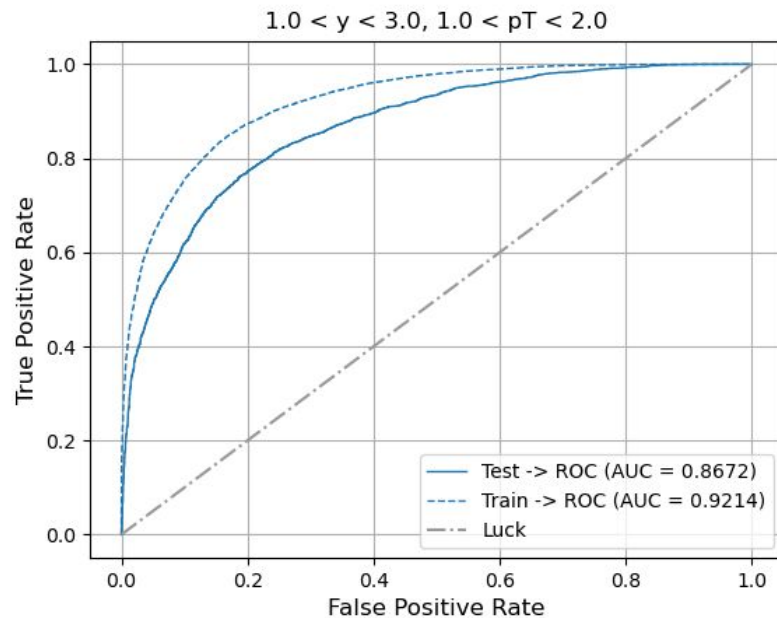
BDT results: with additional hyperparameters

n_estimators: (100, 800) → 728
max_depth: (1, 8) → 8
learning_rate: (0.01, 0.3) → 0.011
min_child_weight: (1, 10) → 10
min_split_loss: (0, 5) → 1



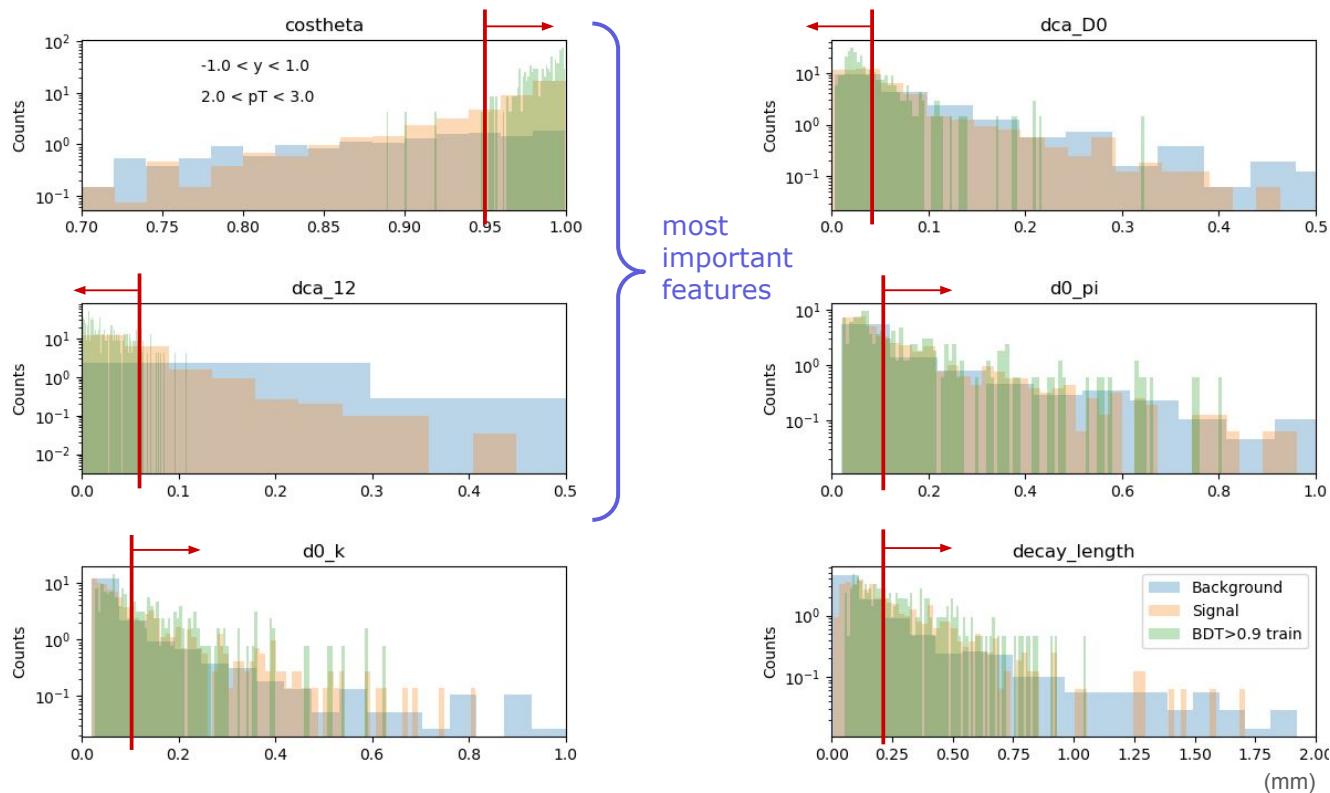
Slight improvement in test ROC AUC, but still overfits.

→ Limit the ranges of n_estimators and max_depth, or increase values of regulators.



BDT results vs. box cuts

BDT vs. box cuts: topological selections



Box cuts from STAR D0 analysis
 $(-1 < y < 1, 2 < p_T < 3)$:

0 - 10% p_T (GeV/c)	(2,3)
Decay Length (μm)	> 232
DCA ₁₂ (μm)	< 63
DCA _{D0} (μm)	< 40
DCA _{π} (μm)	> 97
DCA _K (μm)	> 94
$\cos(\theta)$	> 0.95

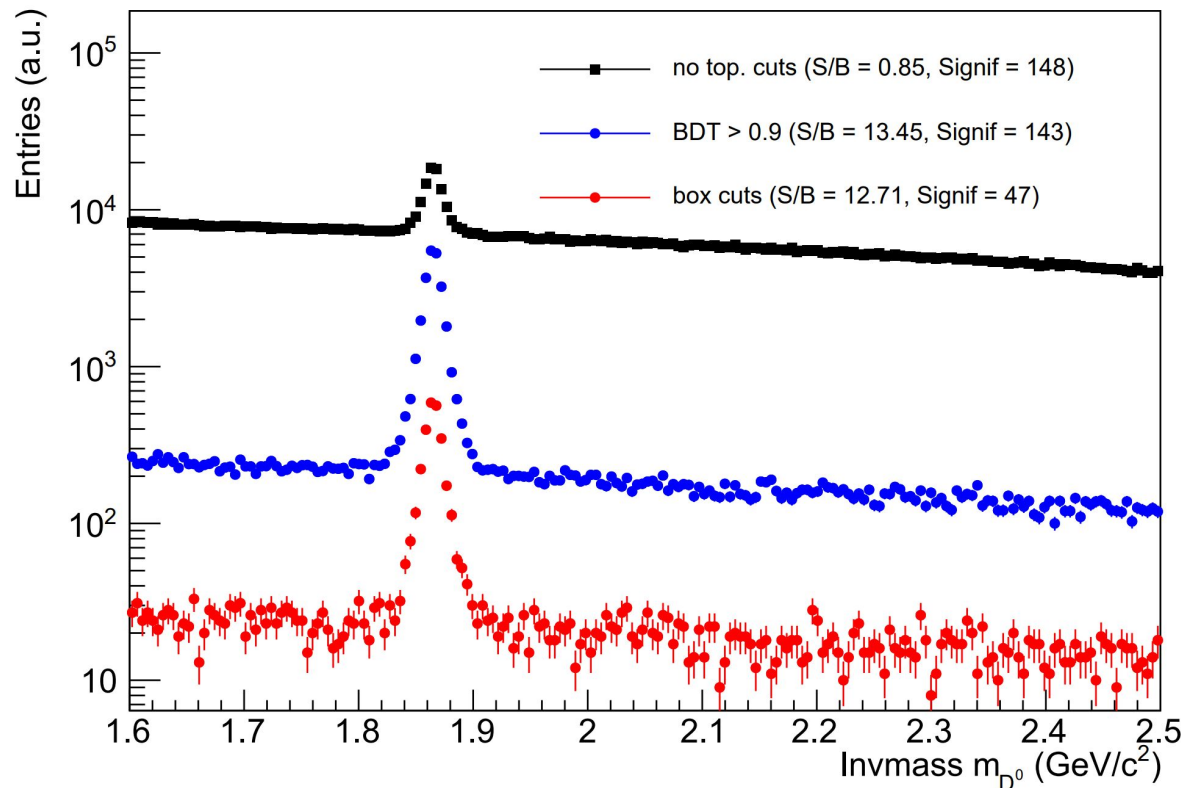
Phys. Rev. C 99, 034908 (2019)

BDT selections are looser
 and more flexible than
 box cuts.

Similar treatments of the
 most important features.

BDT vs. box cuts: D0 invariant mass

$-1 < y < 1, 2 < p_T < 3$



BDT yields higher Signif

- Similar S/B
- Higher statistics

Conclusions

Summary and future directions

- BDT performance varies in different y and p_T slices
 - Separately train BDT for each slice since topological distributions differ
- Feature selection may not significantly impact BDT
 - More relevant for physical and practical considerations
- Greater flexibility in hyperparameter tuning can lead to overfitting
 - Restrict `max_depth` and `n_estimators` to small values
- BDT is similar to but more efficient than box cuts
 - BDT yields higher significance for similar S/B

Summary and future directions

- BDT performance varies in different y and p_T slices
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- BDT is similar to but more efficient than box cuts
 - BDT yields higher significance for similar S/B

Future directions:

- Optimize analysis for data with machine-background effects
- Further explore hyperparameter optimization
 - Additional regulators (`max_delta_step`, `subsample`)
 - Different sampling and optimization algorithms in Optuna

References

October ep simulations from ePIC

D0 samples (for signal candidates):

/volatile/eic/EPIC/RECO/25.10.4/epic_craterlake/Bkg_Exactly1SignalPer2usFrame/SIDIS/D0_ABCONV/pythia8.306-1.1/10x100/q2_1/hiDiv/

DIS samples (for background candidates):

/volatile/eic/EPIC/RECO/25.10.4/epic_craterlake/Bkg_1SignalPer2usFrame/DIS/NC/10x100/minQ2=1

Frameworks for sample analysis and BDT from Shyam

Analysis framework: https://github.com/eic/snippets/tree/main/JetsAndHF/SecondaryVertex_Chi2

Machine learning framework: https://github.com/eic/snippets/tree/main/JetsAndHF/ML_HF_Reconstruction

Based on hipe4ml: <https://doi.org/10.5281/zenodo.5070131>

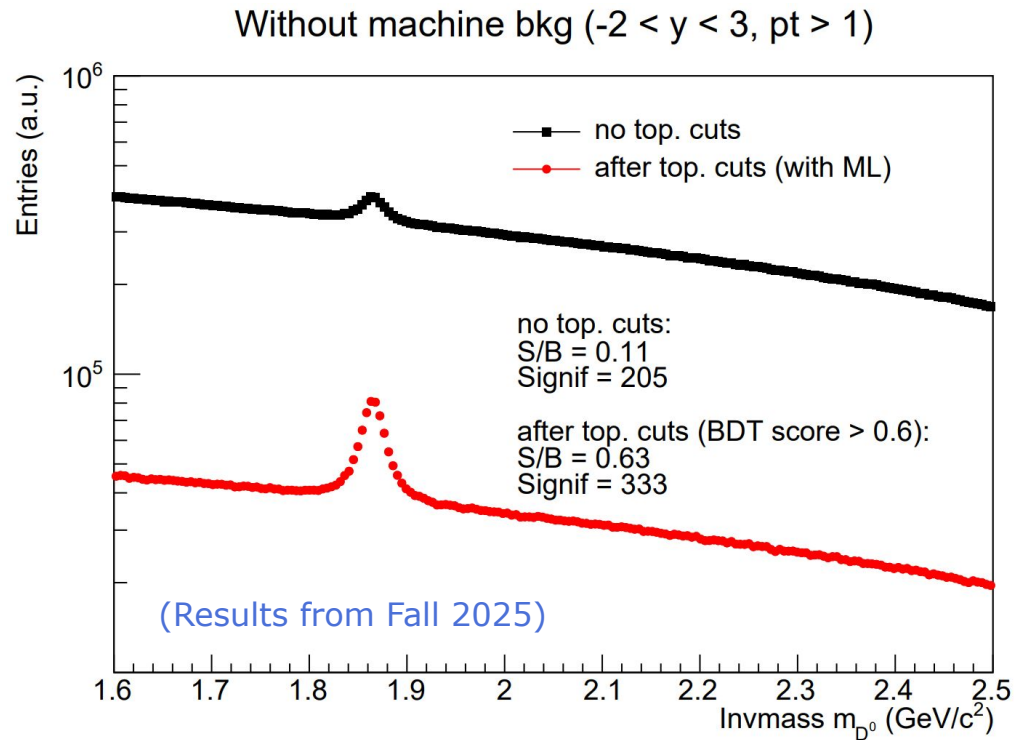
Acknowledgements

Thank you to

- Deepa Thomas
- Shyam Kumar
- Rongrong Ma
- ePIC Collaboration
- RHIP group

Backups

D0 invariant mass reconstruction



$$N_{\text{resampled}} = \frac{N_{\text{candidates}}}{N_{\text{events}}} \times 4.72162e9$$

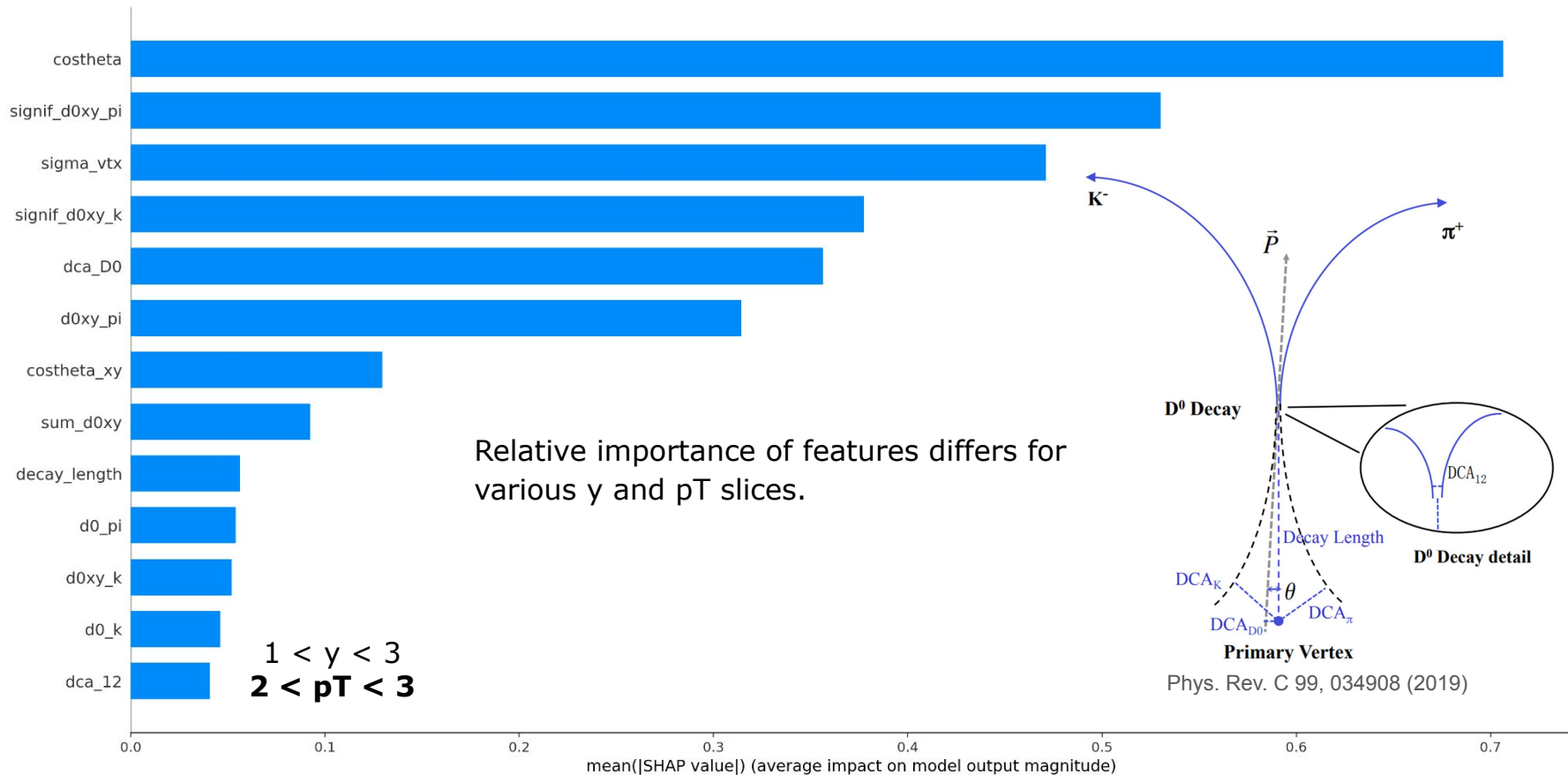
Expected Nevents
for 10 fb⁻¹

$$N_{\text{resampled}}^{\text{ML}} = N_{\text{resampled}} \times \text{eff}(\text{BDT} = 0.6)$$

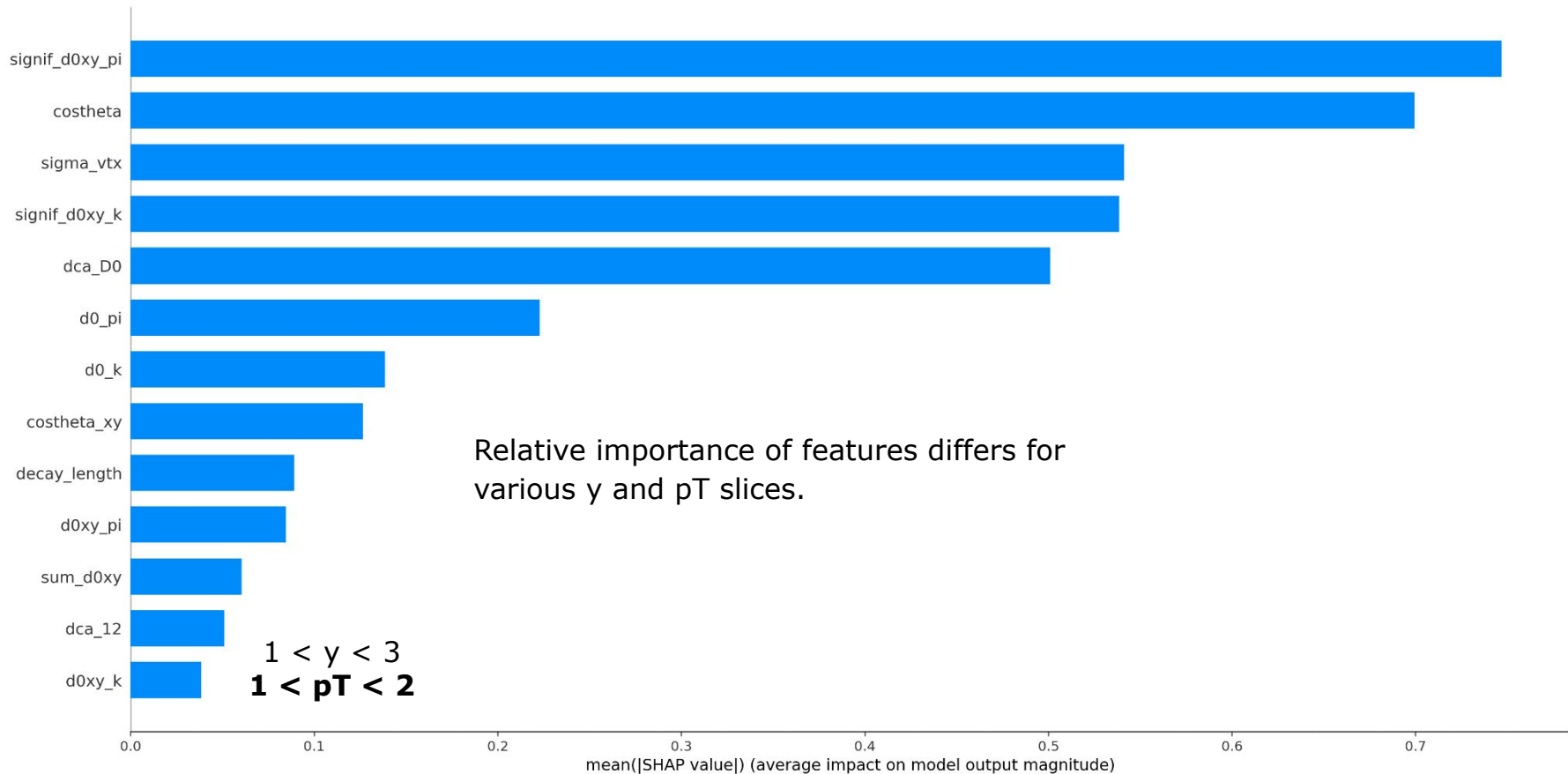
$$S/B = N_{\text{resampled}}^{\text{sig}} / N_{\text{resampled}}^{\text{bkg}}$$

$$\text{Signif} = N_{\text{resampled}}^{\text{sig}} / \sqrt{N_{\text{resampled}}^{\text{sig}} + N_{\text{resampled}}^{\text{bkg}}}$$

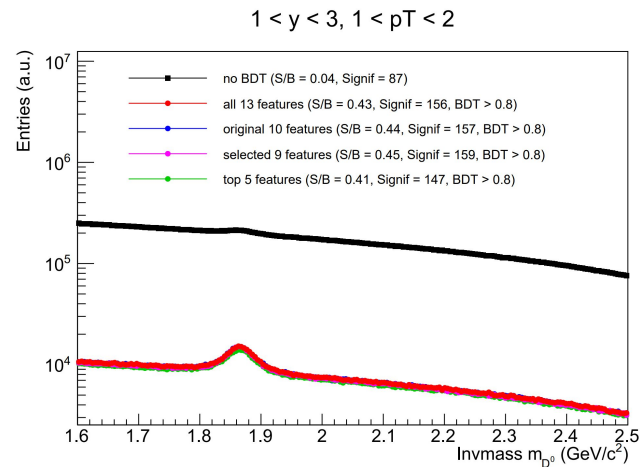
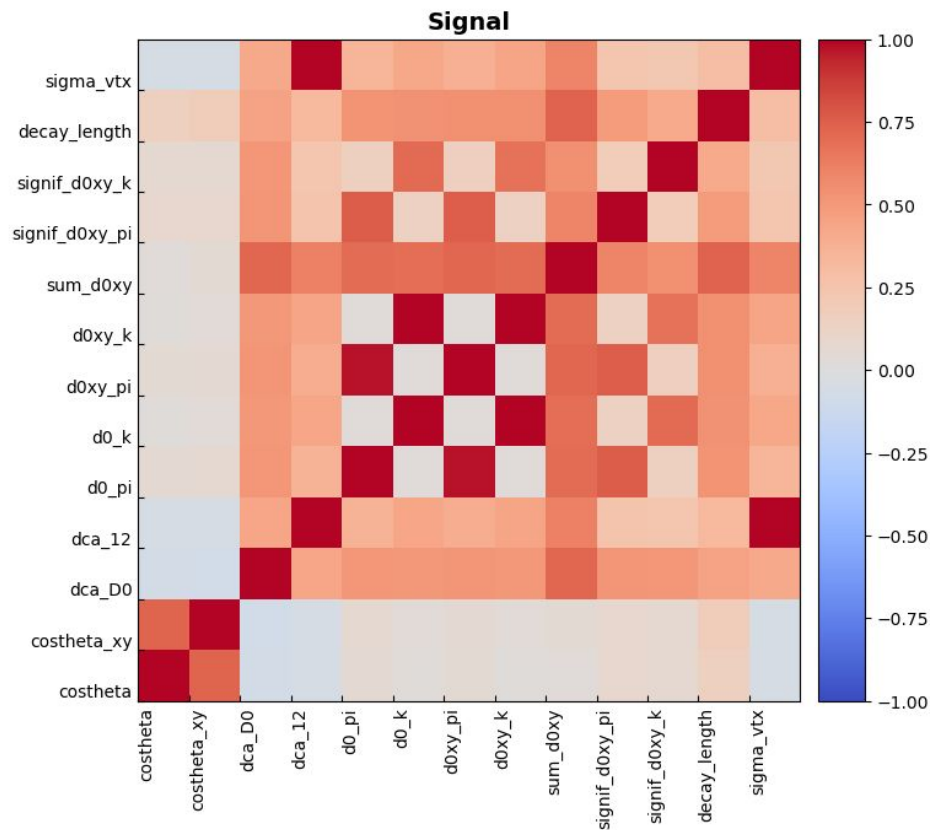
Relative importance of topological features



Relative importance of topological features



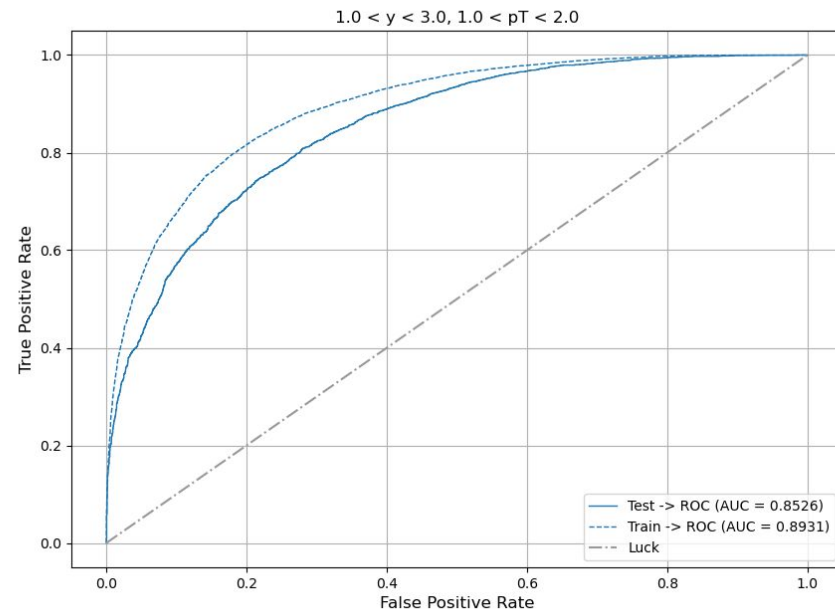
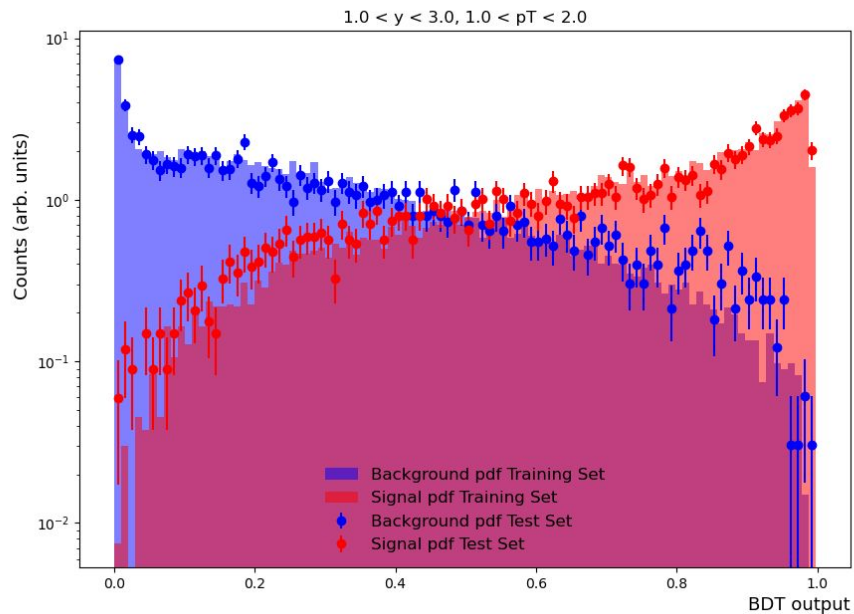
Feature correlations



- Feature correlations do not significantly impact BDT performance.
- Feature selection is more important for physical and practical considerations.
- Correlations also affect SHAP values.

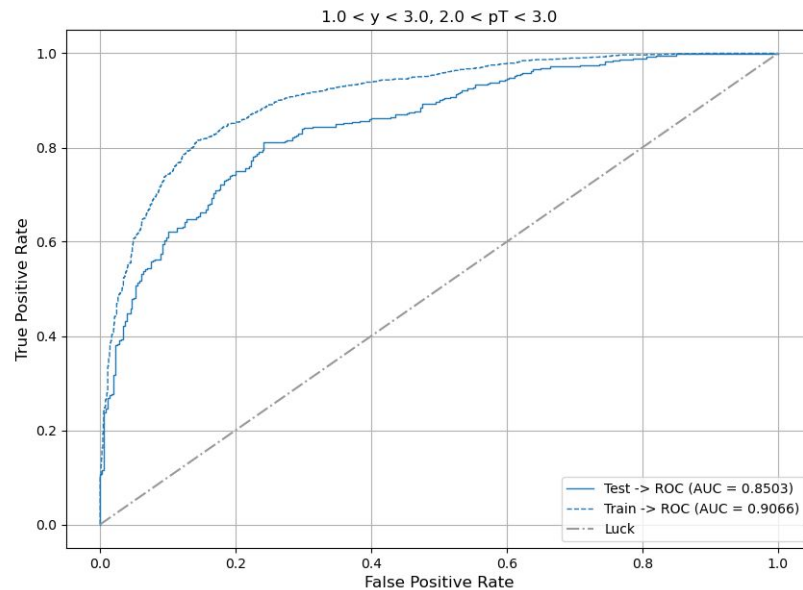
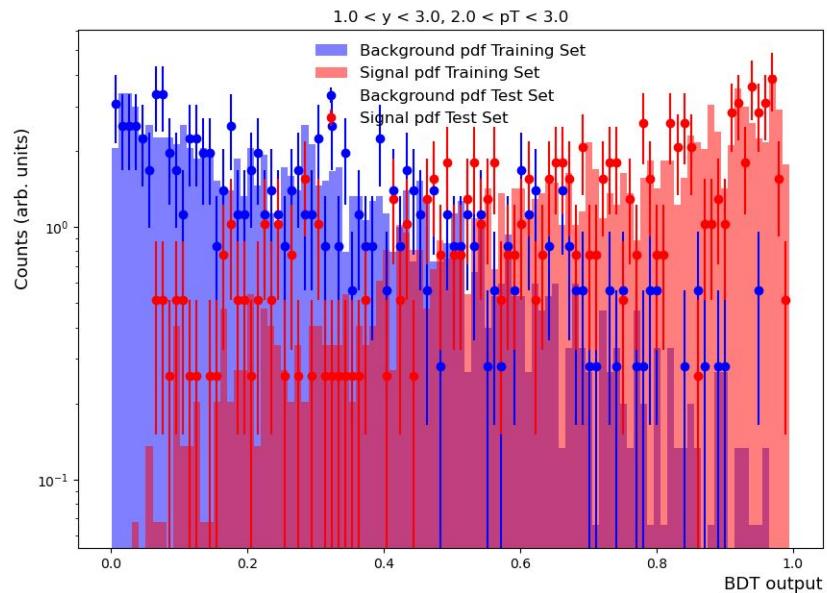
BDT performance: selected 9 features

Relatively stable, without too much overfitting.



BDT performance: all 13 features

Too many features leads to wider BDT output distributions and overfitting.



BDT performance: top 5 features

Too few features also leads to poor ability to discriminate between sig and bkg.

